

Mrudul Mittal

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EDUCATION

Sardar Vallabhbhai National Institute of Technology

BTech in Electronics & VLSI

Current CGPA: 8.23

Surat, Gujarat

2024 – 2028

Agarwal Vidya Vihar

Class XII

Percentage: 93.4%

Surat, Gujarat

2024

SKILLS

Languages: Python, C, C++, PyTorch, SQL

Deep/Machine Learning: RAG Pipelines, LLMs, GANs, CNNs, Transformers, ViTs, XGBoost

Libraries/Tools: LangChain, NumPy, Pandas, Matplotlib, OpenCV, Git, MATLAB

Edge AI & Hardware: Arduino, Embedded C, Raspberry Pi, Jetson Nano, StereoCamera Interfacing

EXPERIENCE

Research Intern | IIT Bombay

Present

3D Botanical Point Cloud Denoising and Reconstruction

- Developing a 3D reconstruction and denoising pipeline for Arabidopsis plant scans to support botanical research and geometric analysis.
- Processing complex point cloud datasets and configuring environments for high-fidelity 3D modeling and data acquisition.

Computer Vision Engineer | Team Drishti (Robocon 2026)

- Architecting comprehensive visual systems and robust object detection pipelines to facilitate autonomous perception in dynamic arena environments.
- Implementing depth estimation algorithms and processing visual data to enable accurate 3D spatial awareness and precise positioning for competitive robots.

PROJECTS & COMPETITIONS

YouTube RAG Chatbot

- Engineered a Retrieval-Augmented Generation (RAG) pipeline utilizing LangChain and the YouTube Transcript API to extract and process video transcripts for interactive querying.
- Integrated the Groq API and open-source Hugging Face models to deploy Large Language Models (LLMs) efficiently, bypassing standard paid API constraints.
- Implemented optimized text chunking and vector retrieval strategies to ensure low-latency document search and accurate, context-aware responses.

Deep Image Restoration | NTIRE 2026 Challenge (CVPR 2026)

- Engineered a dual-branch neural network for high-resolution restoration, mitigating training instabilities against FFT overflows and optimizing multi-scale feature fusion.
- Secured an 8th-place overall ranking in the Image Shadow Removal Challenge by fine-tuning hybrid loss landscapes (Charbonnier + VGG).
- Achieved superior perceptual quality in reconstructed outputs, evaluated by competitive LPIPS of 0.079 and FID of 33.27.

Power Grid Energy Forecasting Dashboard | PowerMind Hackathon

- Architected a multi-strategy time series forecasting pipeline utilizing an ensemble of XGBoost, LightGBM, and CatBoost with weighted blending.
- Integrated Explainable AI (SHAP and LIME) to ensure model transparency, alongside a residual corrector to mitigate prediction bias.
- Developed a full-stack dashboard for live simulation and dynamic energy consumption tracking under real-world constraints.